



Artificial Intelligence (SECJ3553)

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Section 03

Theme: Smart Education

Task: Proposal

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Problem Statement

In the current Malaysian education system, students often face difficulties in connecting what they have learned in the lesson to the real-world applications. Many students struggle to understand how their learning can translate into practical industry tasks when academic learning focuses on theoretical knowledge rather than hands-on ability. This gap between theory and practice leads to reduced confidence and preparedness for students when entering the workforce. They may face early career rejections for these very reasons.

Moreover, it is extremely challenging to assign relevant real-world projects that highly match students' skill levels. The majority of online projects are either too advanced or unrelated to what students have learned in university. Therefore, they hardly practice their academic knowledge.

Overall, deficiencies in real-world experience limits students' ability to develop critical skills and delay their preparation for the future. Hence, we need an AI-driven platform that links classroom lessons to actual industry projects. By offering level-appropriate and task-based modules, the platform can strongly build students' confidence, develop their ability, and cultivate a problem-solving mindset that is suitable for future careers.

AI Solutions

There are three core AI applications in the proposed system. The first AI component is the Intelligent Project Matching system. This system allows students to input what they learnt or their skills, like "I can build a simple website that connects to a database." This input will then be analyzed by AI to understand the meaning of their skills and current skill level. Then it allows the AI to interpret the student's current abilities and match them with the most relevant simulated project scenario from the system's library.

The second AI component is the Guided Task Breakdown system which focuses on streamlining the learning process and navigation within a project. To make complex projects become manageable, the system uses AI to create a personalized path for the student. Each project is built like a roadmap with many smaller, level-based steps with prerequisites, then the AI navigates this roadmap to provide the student with the correct task, and ensure they follow a logical workflow that matches their skills and are never given a task that they are not familiar with. This process simulates the dependencies and structure of a real-world project, allowing students to be more familiar with company projects.

The last AI component is Career Suggestion System. This system can observe and analyze the student's performance and task preferences throughout the simulations. By using the built-in logic, the AI will evaluate the patterns and provide personalized career suggestions like "Your work on this project shows a strong aptitude for the database administrator role." This will help students to understand how their practical skills can be applied in their future career.

Goals of AI Solutions

Our AI solution is designed to accomplish these three main goals:

1. **Match Skills to Projects:** To "connect the dots" between what students learn in class and real-world projects, so that they can see how their knowledge is applied in their future job.
2. **Guide Students Step-by-Step:** To act as a smart project manager that breaks down big and complex projects into small and manageable tasks, then telling the student what to do next, so that they do not get overwhelmed.
3. **Suggest Career Paths:** To act like a helpful mentor by observing the performances and potential skills of the students, then suggest future job roles that match their skills.

Process of Empathize in DT



Process of Define in DT

In the Define stage, we analyze insights gathered from understanding university students to create a clear, human-centered problem definition. This phase ensures we focus on solving what university students need and why they struggle to apply academic learning in real-world contexts.

User	Needs	Insights	Problems Statement	Design Criteria / Notes
University students that are currently enrolled in degree programs who are eager to gain practical skills alongside their academic studies.	Need 1: Students need opportunities to apply what they learn in class to real-world problems through practical experiences. Need 2: Students need a system that matches their knowledge and skill level with suitable, industry-based projects. Need 3: Students need to understand how each academic topic connects to real industry situations and job roles. Need 4: Students need a safe, simulated environment to practice and build confidence before entering the workforce.	Insight 1: Students often forget or lose interest in what they learn because they can't see how classroom lessons connect to real-life work situations. Insight 2: When projects are too hard, students feel overwhelmed; when they're too easy, they feel bored — both make learning less effective. Insight 3: Students feel uncertain about their future careers because they can't picture how their current subjects are used in real jobs. Insight 4: Many students fear failure in real workplaces because they've never had a chance to practice in a realistic but risk-free environment.	University students need a smart, AI-based learning platform that connects what they learn in class with real-world projects and provides step-by-step guidance and career suggestions because they struggle to apply theory to practice and understand how their skills relate to future careers.	- The platform should use AI to match students' learning topics and skills with suitable projects. - It should break down complex projects into small, easy-to-follow tasks. - It should offer personalized career suggestions based on student performance and preferences. - It should fit smoothly with classroom learning and not create extra workload. - It should help students build confidence and be ready for real-world jobs.